

Object Oriented Programming | Java Operator

An **operator** is a symbol used to perform an operation on values or variables.

Example: `+`, `-`, `*`, `/`, `==`

Types of Operators in Java

1. Arithmetic Operators

Used to perform **mathematical calculations**.

Example: `+`, `-`, `*`, `/`, `%`

2. Relational Operators

Used to **compare two values**.

Example: `>`, `<`, `==`, `!=`, `>=`, `<=`

3. Bitwise Operators

Used to perform operations on **individual bits** of values.

Example: `&`, `|`, `^`, `~`

4. Boolean Logical Operators

Used to perform **logical operations** on boolean values.

Example: `&&`, `||`, `!`

5. Conditional Operator (`?:`)

Used to make a **simple decision between two values**.

Example:

```
age >= 18 ? "Adult" : "Minor"
```

Conditional Operator (`?:`)

The **conditional operator** is a ternary operator used to choose **one of two values based on a condition**.

Syntax

```
condition ? value1 : value2;
```

- If the condition is **true**, `value1` is selected.
- If the condition is **false**, `value2` is selected.

Example

```
int age = 20;

String result = age ≥ 18 ? "Adult" : "Minor";

System.out.println(result);
```

Output

Adult

Explanation

Here, `age ≥ 18` is **true**, so `"Adult"` is selected.

Question

Write a Java program to demonstrate the use of the Conditional Operator (`?:`).

Answer

```
public class Test {

    public static void main(String args[]) {

        int a, b;

        a = 10;

        b = (a == 1) ? 20 : 30;

        System.out.println("Value of b is : " + b);

        b = (a == 10) ? 20 : 30;

        System.out.println("Value of b is : " + b);

    }

}
```

Output

```
Value of b is : 30  
Value of b is : 20
```

Explanation

```
b = (a == 1) ? 20 : 30;
```

`a == 1` is **false**, so `b = 30`.

```
b = (a == 10) ? 20 : 30;
```

`a == 10` is **true**, so `b = 20`.

Conditional Operator Format

```
condition ? value_if_true : value_if_false;
```

Control Statements in Java

Control statements are used to **control the flow of execution** of a Java program.

1. Selection Statements

Selection statements are used to **make decisions** in a program.

- **if** — Executes a block of code when a condition is true.
- **switch** — Selects one block of code from multiple choices.

2. Iteration Statements

Iteration statements are used to **repeat a block of code**.

- **for loop** — Repeats code for a specific number of times.
- **while loop** — Repeats code while a condition is true.
- **do-while loop** — Executes the code at least once, then repeats while the condition is true.

3. Jump Statements

Jump statements are used to **change the normal flow of execution**.

- **break** — Stops the loop or switch statement.
- **continue** — Skips the current iteration and moves to the next iteration.
- **return** — Exits from a method and can return a value.

1. Check Whether a Number is Even or Odd

Question

Write a Java program to check whether a number is even or odd.

Answer

```
public class EvenOdd {  
  
    public static void main(String[] args) {  
  
        int num = 10;  
  
        if (num % 2 == 0)  
            System.out.println("Even");  
        else  
            System.out.println("Odd");  
    }  
}
```

Output

Even

2. Check Whether a Number is Positive or Negative

Question

Write a Java program to check whether a number is positive or negative.

Answer

```
public class PositiveNegative {  
  
    public static void main(String[] args) {  
  
        int num = 10;  
  
        if (num ≥ 0)  
            System.out.println("Positive");  
        else  
            System.out.println("Negative");  
    }  
}
```

Output

Positive

3. Check Whether a Year is a Leap Year or Not

Question

Write a Java program to check whether a year is a leap year or not.

Answer

```
public class LeapYear {  
  
    public static void main(String[] args) {  
  
        int year = 2024;  
  
        if (year % 4 == 0)  
            System.out.println("Leap Year");  
        else  
            System.out.println("Not a Leap Year");  
    }  
}
```

Output

```
Leap Year
```

4. Find the Maximum Value of Three Numbers

Question

Write a Java program to find the maximum value among three numbers.

Answer

```
public class Maximum {  
  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 20;  
        int c = 15;  
  
        if (a > b && a > c)  
            System.out.println(a + " is maximum");  
        else if (b > c)  
            System.out.println(b + " is maximum");  
        else  
            System.out.println(c + " is maximum");  
    }  
}
```

Output

```
20 is maximum
```